RV College of Engineering®

Go, Change the World

Academic Year 2023-24 (EVEN Semester)

USN	1	R	V							

Department of Artificial Intelligence and Machine Learning

Course Code:	:	AI244AI	Date	:	19-06-2024
Semester	:	IV	Time	:	12:00 – 1:30 PM
Max Marks	:	50	Duration	:	90 mins

Artificial Intelligence and Machine Learning

CIE 1

Note: Answer all the Questions

SL. No		Questions	M	BT	CO
1	a	A chatbot is an artificial intelligence (AI) software that can simulate a conversation (or a chat) with a user in natural language through messaging applications, websites, mobile apps, or through the telephone. Give a PEAS description of a chatbot that is used to promote cultural events in a college and justify your description.	5	L3	CO1
	b	Mention the types of environments suitable for the following tasks and justify your selection briefly; i. The agent monitors the boiler's temperature in a manufacturing plant. ii. Agent-detecting network attacks	5	L3	CO1
2	a	Differentiate the working of the simple-reflex agents and model-based reflex agents.	5	L2	CO1
	b	By considering any of the domains, such as healthcare, retail, autonomous cars, or others, list four areas for the application of Artificial Intelligence.	5	L3	CO1
3	a	Compare various types of machine learning techniques and their categories with suitable applications.	5	L2	CO1
	b	Illustrate the bias-variance trade-off issue in machine learning. Identify the techniques used to avoid overfitting and underfitting scenarios.	5	L2	CO2
4		Consider the following data: the Y label is whether or not the child goes out to play. Calculate the following:	10	L3	CO5

Day	Weather	Temperature	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Cloudy	Hot	High	Weak	Yes
3	Sunny	Mild	Normal	Strong	Yes
4	Cloudy	Mild	High	Strong	Yes
5	Rainy	Mild	High	Strong	No
6	Rainy	Cool	Normal	Strong	No
7	Rainy	Mild	High	Weak	Yes
8	Sunny	Hot	High	Strong	No
9	Cloudy	Hot	Normal	Weak	Yes
10	Rainy	Mild	High	Strong	No

1. Calculate the IG (information gain) for each attribute (feature)
2. Choose which feature to split with and repeat for each level.
3. Choose a feature for each node to split on

		4. Build Final Tree			
5	a	The following data are recorded while predicting the spam email using a classification model. i. Correct prediction – 0 spam, 75 non-spam ii. Incorrect prediction- 30 spam, 700 non-spam Calculate the Accuracy, sensitivity, precision, and F-measure of the model. Identify the problem with the above classification model.	5	L3	CO5
5	b	Analyze and identify issues in the following machine learning scenarios and propose solutions (s) for them a. A student dataset has an employee age value of 230. b. An object recognition model training set error is 95%, and the test set error is 15%. c. Malignant tumour classification model training set error is 25%, and test set error is 23%.	5	L4	CO1

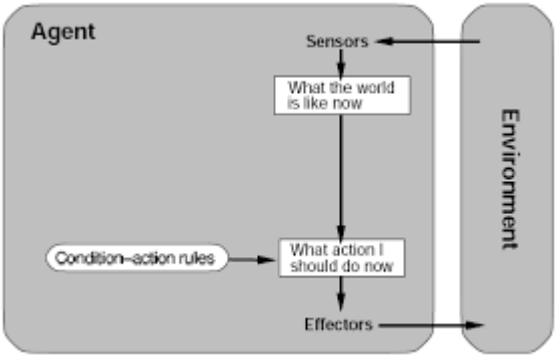
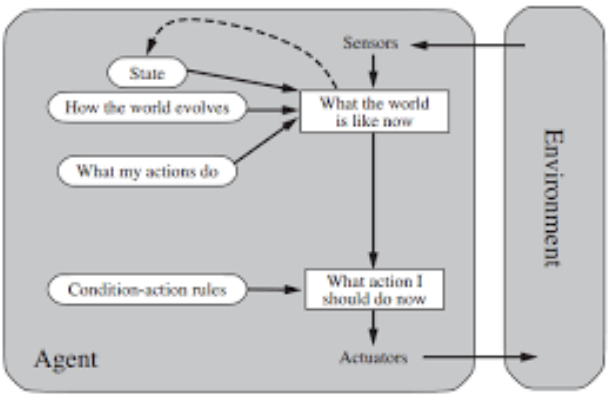
M-Marks, BT-Blooms Taxonomy Levels, CO-Course Outcomes

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks CIE	30	5	-	-	15	-	15	30	5	-	-

Course Outcomes: After completing the course, the students will be able to:-	
CO1	Explain and apply AI & ML algorithms to address various requirements of real-world problems
CO2	Design and develop AI and ML solutions to benefit society, science and industry.
CO3	Use modern tools to create AI and ML solutions.
CO4	Demonstrate effective communication through team presentations and reports to analyze the impact of AI and ML solutions on society and nature.
CO5	Conduct Performance evaluation, modelling and validation of AI and ML solutions benefitting lifelong learning.

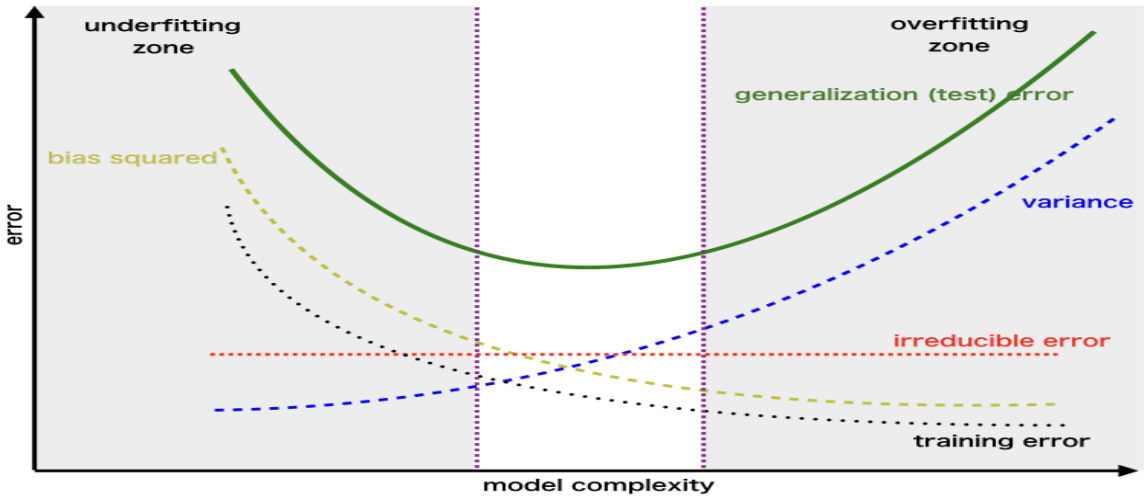
Scheme and Solutions

Q. No	Solutions	M
1a.	PEAS Description of the Chatbot to promote cultural events in the college; SAMPLE ANSWER: Performance: User Engagement, Event Awareness, User Satisfaction, Conversion Rate, Timeliness, etc. (Any TWO – 2M) Environment: Digital Platforms, User location and demographics, Mobile, etc. (Any 1 – 1M) Actuators: Notifications, Advertisements, Messages, etc. (Any 1 – 1M) Sensors: User data, Event data, Data from Social Networks, etc. (Any 1 – 1M)	
1b.	i. **The Agent Monitors the Boiler's Temperature in a Manufacturing Plant (2.5M)** **Suitable Environment: Partially Observable, Dynamic, Continuous, and Deterministic** 1. **Partially Observable:** The agent may not have access to all the environmental variables influencing the boiler's temperature, such as external weather conditions or internal pressure variations, making the environment partially observable. 2. **Dynamic:** The boiler's temperature can change rapidly due to varying operational conditions, heat exchange processes, and potential malfunctions. 3. **Continuous:** The boiler's temperature is a continuous variable that can take on a wide range of values, requiring precise monitoring and control. 4. **Deterministic:** Given the same initial conditions and inputs, the temperature changes predictably based on known physical principles, leading to deterministic behavior. ii. **Agent Detecting Network Attacks** **Suitable Environment: Partially Observable, Dynamic, Discrete, and Stochastic (2.5M)** 1. **Partially Observable**: The agent may not have complete visibility into all network activities or the internal workings of connected devices, making the environment partially observable. 2. **Dynamic**: Network conditions and potential attack vectors can change rapidly as new devices connect, data traffic fluctuates, and attackers use different techniques.	

	3. **Discrete**: Network events, such as packet transfers, connection requests, and alerts, are discrete events that occur at specific times. 4. **Stochastic**: The occurrence of network attacks is probabilistic, with uncertainty in both timing and method, resulting in a stochastic environment	
2a.	Simple Reflex Agent: (Diagram: 1M + Discussion 1.5M)  Model-based Reflex Agent: (Diagram: 1M + Discussion 1.5M)  	
2b.	Each application of AI in a selected domain carries 1.25 M = 1.25 M X 4 EXAMPLE: (Healthcare) 1. To create personalized treatment plans 2. Healthcare data analysis 3. Disease prediction 4. Patient Monitoring	
3a.	**Supervised machine learning** / **Unsupervised machine learning** / **Reinforcement** Used for classifying (classes and labels are given) / To find pattern (No idea about class/labels) / Rewards – if guessed correctly	05

		(No idea about class/labels)
Labelled training data	Unknown and unlabeled data	Model learns and updates itself through rewards/punishment
Classification and regression	Clustering and association analysis	No such types
Simple to understand	More difficult to understand and implement	Most complex
Navie bayes, KNN, LR, SVM etc	Kmeans, PCA, DBSCAN, apriori	Q-Learning, Sarsa
Hand writing recognition, stock market prediction, disease prediction, fraud detection	Market based analysis, recommender system, customer segmentation	Self driving cars, intelligent robots

3b. 05



To avoid overfitting (high variance), try the following –

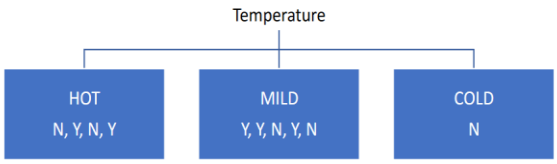
- Increase the training data (collecting more data/augment the training dataset)
- Less complex model
- Early stop (reduce epochs)
- Remove features (not recommended)
- Assembling
- L1/L2 regularization to simplify your model.

To avoid underfitting (high bias):

- Try to increase complexity of model
- Train for longer time
- Try to increase the number of features by finding new features or making new features from the existing ones.
- Decrease regularization
- Use different model

4.	Initial entropy $= H(Y) = -\sum P(Y=y)\log 2\, P(Y=y)y$ $= -P(Y=yes)\log 2\, P(Y=yes) - P(Y=no)\log 2\, P(Y=no)$ $= -(0.5)\log 2(0.5) - (0.5)\log 2(0.5) = 1$ IG = 1-0.7884 = 0.2116	10

Temperature:



Total entropy of this division is:

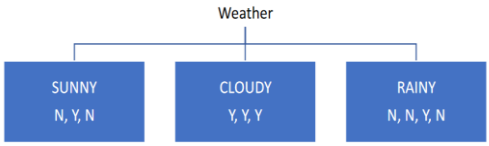
$$H(Y \mid temp) = -\sum_x P(temp = x)\sum_y P(Y=y \mid temp = x)\log_2 P(Y=y \mid temp = x)$$

$$= -(P(temp = H)\textstyle\sum_y P(Y=y \mid temp = H)\log_2 P(Y=y \mid temp = H) + P(temp = M)\sum_y P(Y=y \mid temp = M)\log_2 P(Y=y \mid temp = M) + P(temp = C)\sum_y P(Y=y \mid temp = C)\log_2 P(Y=y \mid temp = C))$$

$$= -((0.4)((\tfrac{1}{2})\log_2(\tfrac{1}{2}) + (\tfrac{1}{2})\log_2(\tfrac{1}{2})) + (0.5)((\tfrac{3}{5})\log_2(\tfrac{3}{5}) + (\tfrac{2}{5})\log_2(\tfrac{2}{5})) + (0.1)((1)\log_2(1) + (0)\log_2(0)))$$

= 0.7884

Weather:



Total entropy of this division is:

$$H(Y \mid weather) = -\sum_x P(weather = x)\sum_y P(Y=y \mid weather = x)\log_2 P(Y=y \mid weather = x)$$

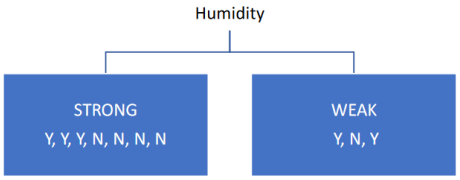
$$= -(P(weather = S)\textstyle\sum_y P(Y=y \mid weather = S)\log_2 P(Y=y \mid weather = S) + P(weather = C)\sum_y P(Y=y \mid weather = C)\log_2 P(Y=y \mid weather = C) + P(weather = R)\sum_y P(Y=y \mid weather = R)\log_2 P(Y=y \mid weather = R))$$

$$= -((0.3)((\tfrac{1}{3})\log_2(\tfrac{1}{3}) + (\tfrac{2}{3})\log_2(\tfrac{2}{3})) + (0.3)((1)\log_2(1) + (0)\log_2(0)) + (0.4)((\tfrac{1}{4})\log_2(\tfrac{1}{4}) + (\tfrac{3}{4})\log_2(\tfrac{3}{4})))$$

= 0.6

IG(Y, weather) = 1 – 0.6 = 0.4

Humidity:



Total entropy of this division is:

$$H(Y \mid hum) = -\sum_x P(hum = x)\sum_y P(Y=y \mid hum = x)\log_2 P(Y=y \mid hum = x)$$

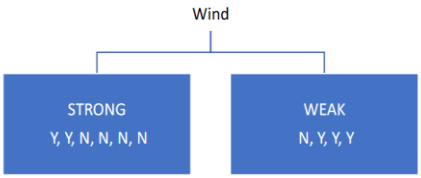
$$= -(P(hum = H)\textstyle\sum_y P(Y=y \mid hum = H)\log_2 P(Y=y \mid hum = H) + P(hum = N)\sum_y P(Y=y \mid hum = N)\log_2 P(Y=y \mid hum = N)$$

$$= -((0.7)((\tfrac{3}{7})\log_2(\tfrac{3}{7}) + (\tfrac{4}{7})\log_2(\tfrac{4}{7})) + (0.3)((\tfrac{2}{3})\log_2(\tfrac{2}{3}) + (\tfrac{1}{3})\log_2(\tfrac{1}{3}))$$

= 0.8651

IG(Y, hum) = 1 – 0.8651 = 0.1349

Wind:



Total entropy of this division is:

$$H(Y \mid wind) = -\sum_x P(wind = x)\sum_y P(Y=y \mid wind = x)\log_2 P(Y=y \mid wind = x)$$

$$= -(P(wind = S)\textstyle\sum_y P(Y=y \mid wind = S)\log_2 P(Y=y \mid wind = S) + P(wind = W)\sum_y P(Y=y \mid wind = W)\log_2 P(Y=y \mid wind = W)$$

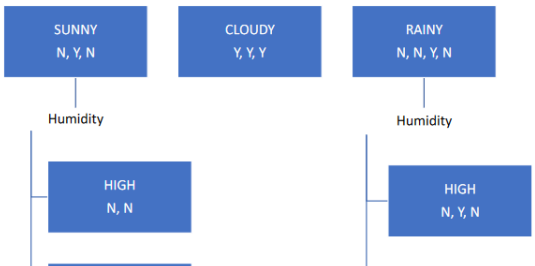
$$= -((0.6)((\tfrac{2}{6})\log_2(\tfrac{2}{6}) + (\tfrac{4}{6})\log_2(\tfrac{4}{6})) + (0.4)((\tfrac{1}{4})\log_2(\tfrac{1}{4}) + (\tfrac{3}{4})\log_2(\tfrac{3}{4}))$$

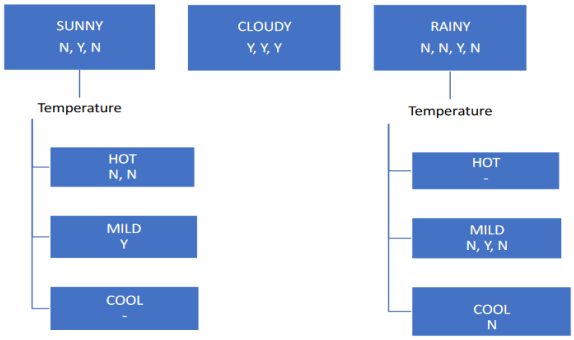
= 0.8755

IG(Y, wind) = 1 – 0.8755 = 0.1245

IG(Y, wind) = 0.1245
IG(Y, hum) = 0.1349
IG(Y, weather) = 0.4
IG(Y, temp) = 0.2116

Humidity





Entropy of "Sunny" node = $-((\frac{1}{3})\log_2(\frac{1}{3}) + (\frac{2}{3})\log_2(\frac{2}{3})) = 0.9183$

Entropy of its children = 0

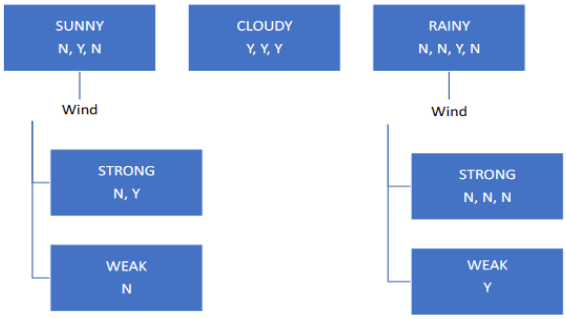
IG = 0.9183

Entropy of "Rainy" node = $-((\frac{1}{4})\log_2(\frac{1}{4}) + (\frac{3}{4})\log_2(\frac{3}{4})) = 0.8113$

Entropy of children = $-(\frac{3}{4})((\frac{1}{3})\log_2(\frac{1}{3}) + (\frac{2}{3})\log_2(\frac{2}{3})) + 0 = 0.6887$

IG = 0.1226

Wind



Entropy of "Sunny" node = $-((\frac{1}{3})\log_2(\frac{1}{3}) + (\frac{2}{3})\log_2(\frac{2}{3})) = 0.9183$

Entropy of its children = $-(\frac{2}{4})((\frac{1}{2})\log_2(\frac{1}{2}) + (\frac{1}{2})\log_2(\frac{1}{2})) + 0 = 0.6667$

IG = 0.2516

Entropy of "Rainy" node = $-((\frac{1}{4})\log_2(\frac{1}{4}) + (\frac{3}{4})\log_2(\frac{3}{4})) = 0.8113$

Entropy of children = 0

IG = 0.8113

"Sunny node":

IG(Y, weather) = IG(humidity) = 0.9183

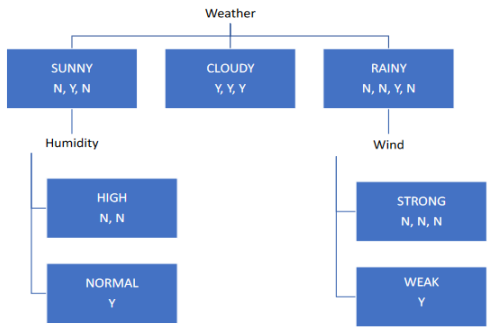
IG(Y, wind) = 0.2516

"Rainy node":

IG(Y, weather) = IG(Y, humidity) = 0.1226

IG(Y, wind) = 0.8113

Final Tree!



5a.	Accuracy=0.8697, sensitivity= 0.0323, precision= 0.0132 and F-measure of the model. Identify the problem with above classification model. It is imbalanced dataset. The model failed to predict positive class instances	05
5b.	A student dataset has an employee age value as 230. Outlier, impute with avg, median or other suitable values. b. An object recognition model training set error is 95% and test set error is 15%. overfitting, decrease model complexity. c. Malignant tumor classification model training set error is 25% and test set error is 23%. underfitting, increase model complexity.	05